KITABU QUEST

S4

Subsidiary Mathematics

Mascot: mountain gorilla
Scene: students solving applied mathematics problems with graphs, models, and a mountain gorilla mascot
CBC aligned draft learner book

Think • Explore • Achieve

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Title Page

KITABU QUEST S4 Subsidiary Mathematics

Rwanda REB-CBC learner book

Mascot: mountain gorilla

This KITABU QUEST book turns the curriculum into short lessons, worked examples, guided activities, practice, review, and home links for Rwanda learners.

How To Use This Book

This book helps S4 learners study Subsidiary Mathematics one teachable topic at a time.

1. Start with the inquiry question and say what you already know.
2. Read the Learn section slowly and mark new words.
3. Try the worked example before reading the full solution.
4. Complete the activity and practice tasks.
5. Correct your answer using the success criteria.
6. Ask for support when your answer is still unclear.

Subject method: Learn through examples, activities, practice, and reflection.

Learning Skills In This Book

Each topic builds knowledge, skill, values, communication, collaboration, critical thinking, creativity, digital awareness, and self-confidence.

Study actively: speak answers aloud, draw when useful, show your working, cite evidence, use local examples, and correct mistakes after feedback.

A strong answer is specific. It names the idea, gives evidence or method, applies it to the task, and checks whether it answers the question.

Table Of Contents

Use this table to move through the book one topic at a time. Each topic has a lesson opener, clear teaching, a model or example, guided work, practice, and checks for understanding.

1. Fundamentals of trigonometry
2. SET OF REAL NUMBERS
3. Linear, Quadratic equations and inequalities
4. Polynomial, Rational and Irrational functions
5. Limits of polynomial, rational and irrational functions
6. Differentiation of polynomials, rational and irrational functions and their applications
7. Vector Space of real numbers
8. Matrices of and determinants of order
9. Measures of dispersion
10. Elementary probability

As you finish each topic, return to the success criteria and correct one answer before moving on.

Fundamentals of trigonometry: Lesson Opener

Learning focus: Fundamentals of trigonometry

Country link: a Rwandan school, market, farming, building, transport, or data problem for Fundamentals of trigonometry.

Progression focus: prepare for assessment and real-world use through extended tasks and correction; show methods, units, checks, and a final explanation.

By the end of this topic, you should be able to:
- use the trigonometric concepts and formulas to solve related problems in Physics, Air navigation, Water navigation, bearings, Surveying,

Inquiry questions:
- How can fundamentals of trigonometry help you solve a real problem in Subsidiary Mathematics?

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Fundamentals of trigonometry: Learn

Fundamentals of trigonometry teaches a mathematical idea by using a clear model, a labelled representation, step-by-step working, and a final check.

Key words:
- Fundamentals of trigonometry
- Fundamentals
- trigonometry
- method
- model
- unit
- answer
- check

Fundamentals of trigonometry: Worked Example

Worked example for Fundamentals of trigonometry: Read the task, choose a representation, solve one step at a time, and check the answer in context.

Model:
1. Name what is known.
2. Choose a diagram, table, equation, or model.
3. Complete the calculation or reasoning.
4. Check the result with estimation or inverse reasoning.

Use a Rwanda classroom, market, garden, transport, or home example.

Fundamentals of trigonometry: Vocabulary And Study Skill

Use these words and study moves before you practise Fundamentals of trigonometry. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:
- Fundamentals of trigonometry
- Fundamentals
- trigonometry
- method
- model
- unit
- answer
- check

Study skill:
1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Fundamentals of trigonometry without a clear example, method, or evidence.

Fundamentals of trigonometry: Guided Activity

Guided activity for Fundamentals of trigonometry:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Fundamentals of trigonometry: Practice And Correction

Practice for Fundamentals of trigonometry:

Main outcome: use the trigonometric concepts and formulas to solve related problems in Physics, Air navigation, Water navigation, bearings, Surveying,.

1. Solve one fundamentals of trigonometry question using labelled steps.
2. Show your method with a drawing, table, number sentence, or diagram.
3. Check your answer and write one sentence explaining it.

Answer check:
Your answer should show a clear model, correct method, correct labels, and a final sentence.

Correction check:
- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to fundamentals of trigonometry and solve it.

Fundamentals of trigonometry: Challenge And Remediation

Use this page after practice.

If you need support:
1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Fundamentals of trigonometry.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:
1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:
- method is clear
- working is shown step by step
- answer is checked in context

Fundamentals of trigonometry: Outcome Check

Outcome check for Fundamentals of trigonometry: Use the trigonometric concepts and formulas to solve related problems in Physics, Air navigation, Water navigation, bearings, Surveying,.

1. Write one clear example from the topic.
2. Solve it with labelled steps.
3. Check the answer in context.

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

SET OF REAL NUMBERS: Lesson Opener

Learning focus: SET OF REAL NUMBERS

Country link: a Rwandan school, market, farming, building, transport, or data problem for SET OF REAL NUMBERS.

Progression focus: prepare for assessment and real-world use through extended tasks and correction; show methods, units, checks, and a final explanation.

By the end of this topic, you should be able to:
- think critically to understand and perform operations on the set of real numbers

Inquiry questions:
- How can set of real numbers help you solve a real problem in Subsidiary Mathematics?

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

SET OF REAL NUMBERS: Learn

SET OF REAL NUMBERS teaches how two quantities change together at the same rate.

Core idea:
- In direct proportion, when one quantity is multiplied, the other quantity is multiplied by the same factor.
- A table helps you compare matching values.
- The unit rate tells the value for one item, one litre, one metre, one hour, or one learner.
- Always check that the relationship is consistent before using proportion.

Key words:
- SET OF REAL NUMBERS
- REAL
- NUMBERS
- method
- model
- unit
- answer
- check

SET OF REAL NUMBERS: Worked Example

Worked example for SET OF REAL NUMBERS: If 3 exercise books cost 900 francs, how much do 5 exercise books cost at the same price?

1. Find the cost of 1 book: 900 / 3 = 300 francs.
2. Multiply by 5 books: 300 x 5 = 1500 francs.
3. Check: more books should cost more money.

Answer: 5 exercise books cost 1500 francs.

SET OF REAL NUMBERS: Vocabulary And Study Skill

Use these words and study moves before you practise SET OF REAL NUMBERS. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:
- SET OF REAL NUMBERS
- REAL
- NUMBERS
- method
- model
- unit
- answer
- check

Study skill:
1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about SET OF REAL NUMBERS without a clear example, method, or evidence.

SET OF REAL NUMBERS: Guided Activity

Guided activity for SET OF REAL NUMBERS:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

SET OF REAL NUMBERS: Practice And Correction

Practice for SET OF REAL NUMBERS:

Main outcome: think critically to understand and perform operations on the set of real numbers.

1. Complete the table: 1 pen = 200 francs, 2 pens = __, 5 pens = __.
2. A cyclist travels 18 km in 2 hours at the same speed. How far in 5 hours?
3. Write the unit rate used in each answer.

Answer check:
Answers: 2 pens = 400 francs; 5 pens = 1000 francs. The cyclist travels 45 km in 5 hours because 18 / 2 = 9 km per hour.

Correction check:
- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to set of real numbers and solve it.

SET OF REAL NUMBERS: Challenge And Remediation

Use this page after practice.

If you need support:
1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from SET OF REAL NUMBERS.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:
1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:
- method is clear
- working is shown step by step
- answer is checked in context

SET OF REAL NUMBERS: Outcome Check

Outcome check for SET OF REAL NUMBERS: Think critically to understand and perform operations on the set of real numbers.

1. A recipe uses 2 cups of flour for 6 cakes. How many cups for 18 cakes?
2. Write the unit rate or scale factor.
3. Check that both quantities changed by the same factor.

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Linear, Quadratic equations and inequalities: Lesson Opener

Learning focus: Linear, Quadratic equations and inequalities

Country link: a Rwandan school, market, farming, building, transport, or data problem for Linear, Quadratic equations and inequalities.

Progression focus: prepare for assessment and real-world use through extended tasks and correction; show methods, units, checks, and a final explanation.

By the end of this topic, you should be able to:
- model and solve algebraically or graphically daily life problems using linear, quadratic equations or inequalities

Inquiry questions:
- How can linear, quadratic equations and inequalities help you solve a real problem in Subsidiary Mathematics?

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Linear, Quadratic equations and inequalities: Learn

Linear, Quadratic equations and inequalities teaches how lines and angles describe shapes and directions.

Core idea:
- A line can be straight, parallel, perpendicular, horizontal, vertical, or slanting.
- An angle is a turn between two lines.
- Draw and label diagrams carefully before measuring or classifying.

Key words:
- Linear, Quadratic equations and inequalities
- Linear
- Quadratic
- equations
- inequalities
- method
- model
- unit

Linear, Quadratic equations and inequalities: Worked Example

Worked example for Linear, Quadratic equations and inequalities: A shape has one right angle. How can you show it clearly?

1. Draw the two meeting lines.
2. Mark the square corner to show a right angle.
3. Label it 90 degrees.
4. Name a classroom example, such as a book corner.

Answer: A right angle is a quarter turn, 90 degrees.

Linear, Quadratic equations and inequalities: Vocabulary And Study Skill

Use these words and study moves before you practise Linear, Quadratic equations and inequalities. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:
- Linear, Quadratic equations and inequalities
- Linear
- Quadratic
- equations
- inequalities
- method
- model
- unit

Study skill:
1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Linear, Quadratic equations and inequalities without a clear example, method, or evidence.

Linear, Quadratic equations and inequalities: Guided Activity

Guided activity for Linear, Quadratic equations and inequalities:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Linear, Quadratic equations and inequalities: Practice And Correction

Practice for Linear, Quadratic equations and inequalities:

Main outcome: model and solve algebraically or graphically daily life problems using linear, quadratic equations or inequalities.

1. Draw one right angle and label it 90 degrees.
2. Name one pair of parallel lines in the classroom.
3. Use a ruler to draw a straight line segment and label its endpoints.

Answer check:
Answer check: the right angle should show a square corner; parallel lines stay the same distance apart.

Correction check:
- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to linear, quadratic equations and inequalities and solve it.

Linear, Quadratic equations and inequalities: Challenge And Remediation

Use this page after practice.

If you need support:
1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Linear, Quadratic equations and inequalities.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:
1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:
- method is clear
- working is shown step by step
- answer is checked in context

Linear, Quadratic equations and inequalities: Outcome Check

Outcome check for Linear, Quadratic equations and inequalities: Model and solve algebraically or graphically daily life problems using linear, quadratic equations or inequalities.

1. Draw and label one right angle.
2. Name one pair of parallel lines.
3. Explain how you know.

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Polynomial, Rational and Irrational functions: Lesson Opener

Learning focus: Polynomial, Rational and Irrational functions

Country link: a Rwandan school, market, farming, building, transport, or data problem for Polynomial, Rational and Irrational functions.

Progression focus: prepare for assessment and real-world use through extended tasks and correction; show methods, units, checks, and a final explanation.

By the end of this topic, you should be able to:
- use concepts and definitions of Polynomial, Rational and Irrational functions to determine the domain of Polynomial, Rational and Irrational functions and represent them graphically in simple cases…

Inquiry questions:
- How can polynomial, rational and irrational functions help you solve a real problem in Subsidiary Mathematics?

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Polynomial, Rational and Irrational functions: Learn

Polynomial, Rational and Irrational functions teaches how two quantities change together at the same rate.

Core idea:
- In direct proportion, when one quantity is multiplied, the other quantity is multiplied by the same factor.
- A table helps you compare matching values.
- The unit rate tells the value for one item, one litre, one metre, one hour, or one learner.
- Always check that the relationship is consistent before using proportion.

Key words:
- Polynomial, Rational and Irrational functions
- Polynomial
- Rational
- Irrational
- functions
- method
- model

Polynomial, Rational and Irrational functions: Worked Example

Worked example for Polynomial, Rational and Irrational functions: If 3 exercise books cost 900 francs, how much do 5 exercise books cost at the same price?

1. Find the cost of 1 book: 900 / 3 = 300 francs.
2. Multiply by 5 books: 300 x 5 = 1500 francs.
3. Check: more books should cost more money.

Answer: 5 exercise books cost 1500 francs.

Polynomial, Rational and Irrational functions: Vocabulary And Study Skill

Use these words and study moves before you practise Polynomial, Rational and Irrational functions. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:
- Polynomial, Rational and Irrational functions
- Polynomial
- Rational
- Irrational
- functions
- method
- model
- unit

Study skill:
1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Polynomial, Rational and Irrational functions without a clear example, method, or evidence.

Polynomial, Rational and Irrational functions: Guided Activity

Guided activity for Polynomial, Rational and Irrational functions:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Polynomial, Rational and Irrational functions: Practice And Correction

Practice for Polynomial, Rational and Irrational functions:

Main outcome: use concepts and definitions of Polynomial, Rational and Irrational functions to determine the domain of Polynomial, Rational and Irrational functions and represent them graphically in simple cases….

1. Complete the table: 1 pen = 200 francs, 2 pens = __, 5 pens = __.
2. A cyclist travels 18 km in 2 hours at the same speed. How far in 5 hours?
3. Write the unit rate used in each answer.

Answer check:
Answers: 2 pens = 400 francs; 5 pens = 1000 francs. The cyclist travels 45 km in 5 hours because 18 / 2 = 9 km per hour.

Correction check:
- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to polynomial, rational and irrational functions and solve it.

Polynomial, Rational and Irrational functions: Challenge And Remediation

Use this page after practice.

If you need support:
1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Polynomial, Rational and Irrational functions.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:
1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:
- method is clear
- working is shown step by step
- answer is checked in context

Polynomial, Rational and Irrational functions: Outcome Check

Outcome check for Polynomial, Rational and Irrational functions: Use concepts and definitions of Polynomial, Rational and Irrational functions to determine the domain of Polynomial, Rational and Irrational functions and represent them graphically in simple cases….

1. A recipe uses 2 cups of flour for 6 cakes. How many cups for 18 cakes?
2. Write the unit rate or scale factor.
3. Check that both quantities changed by the same factor.

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Limits of polynomial, rational and irrational functions: Lesson Opener

Learning focus: Limits of polynomial, rational and irrational functions

Country link: a Rwandan school, market, farming, building, transport, or data problem for Limits of polynomial, rational and irrational functions.

Progression focus: prepare for assessment and real-world use through extended tasks and correction; show methods, units, checks, and a final explanation.

By the end of this topic, you should be able to:
- evaluate correctly limits of functionsand apply them to solve related problems

Inquiry questions:
- How can limits of polynomial, rational and irrational functions help you solve a real problem in Subsidiary Mathematics?

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Limits of polynomial, rational and irrational functions: Learn

Limits of polynomial, rational and irrational functions teaches how two quantities change together at the same rate.

Core idea:
- In direct proportion, when one quantity is multiplied, the other quantity is multiplied by the same factor.
- A table helps you compare matching values.
- The unit rate tells the value for one item, one litre, one metre, one hour, or one learner.
- Always check that the relationship is consistent before using proportion.

Key words:
- Limits of polynomial, rational and irrational functions
- Limits
- polynomial
- rational
- irrational
- functions
- method
- model

Limits of polynomial, rational and irrational functions: Worked Example

Worked example for Limits of polynomial, rational and irrational functions: If 3 exercise books cost 900 francs, how much do 5 exercise books cost at the same price?

1. Find the cost of 1 book: 900 / 3 = 300 francs.
2. Multiply by 5 books: 300 x 5 = 1500 francs.
3. Check: more books should cost more money.

Answer: 5 exercise books cost 1500 francs.

Limits of polynomial, rational and irrational functions: Vocabulary And Study Skill

Use these words and study moves before you practise Limits of polynomial, rational and irrational functions. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:
- Limits of polynomial, rational and irrational functions
- Limits
- polynomial
- rational
- irrational
- functions
- method
- model

Study skill:
1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Limits of polynomial, rational and irrational functions without a clear example, method, or evidence.

Limits of polynomial, rational and irrational functions: Guided Activity

Guided activity for Limits of polynomial, rational and irrational functions:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Limits of polynomial, rational and irrational functions: Practice And Correction

Practice for Limits of polynomial, rational and irrational functions:

Main outcome: evaluate correctly limits of functionsand apply them to solve related problems.

1. Complete the table: 1 pen = 200 francs, 2 pens = __, 5 pens = __.
2. A cyclist travels 18 km in 2 hours at the same speed. How far in 5 hours?
3. Write the unit rate used in each answer.

Answer check:
Answers: 2 pens = 400 francs; 5 pens = 1000 francs. The cyclist travels 45 km in 5 hours because 18 / 2 = 9 km per hour.

Correction check:
- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to limits of polynomial, rational and irrational functions and solve it.

Limits of polynomial, rational and irrational functions: Challenge And Remediation

Use this page after practice.

If you need support:
1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Limits of polynomial, rational and irrational functions.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:
1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:
- method is clear
- working is shown step by step
- answer is checked in context

Limits of polynomial, rational and irrational functions: Outcome Check

Outcome check for Limits of polynomial, rational and irrational functions: Evaluate correctly limits of functionsand apply them to solve related problems.

1. A recipe uses 2 cups of flour for 6 cakes. How many cups for 18 cakes?
2. Write the unit rate or scale factor.
3. Check that both quantities changed by the same factor.

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Differentiation of polynomials, rational and irrational functions and their applications: Lesson Opener

Learning focus: Differentiation of polynomials, rational and irrational functions and their applications

Country link: a Rwandan school, market, farming, building, transport, or data problem for Differentiation of polynomials, rational and irrational functions and their applications.

Progression focus: prepare for assessment and real-world use through extended tasks and correction; show methods, units, checks, and a final explanation.

By the end of this topic, you should be able to:
- use the gradient of a straight line as a measure of rate of change and apply this to line tangent and normal of curves in various contexts and use these concepts of differentiation to solve and interpret related rates and optimization problems in various conte
- equation of the tangent to a curve

Inquiry questions:
- How can differentiation of polynomials, rational and irrational functions and their applications help you solve a real problem in Subsidiary Mathematics?

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Differentiation of polynomials, rational and irrational functions and their applications: Learn

Differentiation of polynomials, rational and irrational functions and their applications teaches how two quantities change together at the same rate.

Core idea:
- In direct proportion, when one quantity is multiplied, the other quantity is multiplied by the same factor.
- A table helps you compare matching values.
- The unit rate tells the value for one item, one litre, one metre, one hour, or one learner.
- Always check that the relationship is consistent before using proportion.

Key words:
- Differentiation of polynomials, rational and irrational functions and their applications
- Differentiation
- polynomials
- rational
- irrational
- functions
- their
- applications

Differentiation of polynomials, rational and irrational functions and their applications: Worked Example

Worked example for Differentiation of polynomials, rational and irrational functions and their applications: If 3 exercise books cost 900 francs, how much do 5 exercise books cost at the same price?

1. Find the cost of 1 book: 900 / 3 = 300 francs.
2. Multiply by 5 books: 300 x 5 = 1500 francs.
3. Check: more books should cost more money.

Answer: 5 exercise books cost 1500 francs.

Differentiation of polynomials, rational and irrational functions and their applications: Vocabulary And Study Skill

Use these words and study moves before you practise Differentiation of polynomials, rational and irrational functions and their applications. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:
- Differentiation of polynomials, rational and irrational functions and their applications
- Differentiation
- polynomials
- rational
- irrational
- functions
- their
- applications

Study skill:
1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Differentiation of polynomials, rational and irrational functions and their applications without a clear example, method, or evidence.

Differentiation of polynomials, rational and irrational functions and their applications: Guided Activity

Guided activity for Differentiation of polynomials, rational and irrational functions and their applications:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Differentiation of polynomials, rational and irrational functions and their applications: Practice And Correction

Practice for Differentiation of polynomials, rational and irrational functions and their applications:

Main outcome: use the gradient of a straight line as a measure of rate of change and apply this to line tangent and normal of curves in various contexts and use these concepts of differentiation to solve and interpret related rates and optimization problems in various conte.

1. Complete the table: 1 pen = 200 francs, 2 pens = __, 5 pens = __.
2. A cyclist travels 18 km in 2 hours at the same speed. How far in 5 hours?
3. Write the unit rate used in each answer.

Answer check:
Answers: 2 pens = 400 francs; 5 pens = 1000 francs. The cyclist travels 45 km in 5 hours because 18 / 2 = 9 km per hour.

Correction check:
- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to differentiation of polynomials, rational and irrational functions and their applications and solve it.

Differentiation of polynomials, rational and irrational functions and their applications: Challenge And Remediation

Use this page after practice.

If you need support:
1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Differentiation of polynomials, rational and irrational functions and their applications.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:
1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:
- method is clear
- working is shown step by step
- answer is checked in context

Differentiation of polynomials, rational and irrational functions and their applications: Outcome Check

Outcome check for Differentiation of polynomials, rational and irrational functions and their applications: Use the gradient of a straight line as a measure of rate of change and apply this to line tangent and normal of curves in various contexts and use these concepts of differentiation to solve and interpret related rates and optimization problems in various conte.

1. A recipe uses 2 cups of flour for 6 cakes. How many cups for 18 cakes?
2. Write the unit rate or scale factor.
3. Check that both quantities changed by the same factor.

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Differentiation of polynomials, rational and irrational functions and their applications: Outcome Check

Outcome check for Differentiation of polynomials, rational and irrational functions and their applications: Equation of the tangent to a curve.

1. A recipe uses 2 cups of flour for 6 cakes. How many cups for 18 cakes?
2. Write the unit rate or scale factor.
3. Check that both quantities changed by the same factor.

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Vector Space of real numbers: Lesson Opener

Learning focus: Vector Space of real numbers

Country link: a Rwandan school, market, farming, building, transport, or data problem for Vector Space of real numbers.

Progression focus: prepare for assessment and real-world use through extended tasks and correction; show methods, units, checks, and a final explanation.

By the end of this topic, you should be able to:
- use concepts of vectors in 2D to solve related problems such as distance, angles,…

Inquiry questions:
- How can vector space of real numbers help you solve a real problem in Subsidiary Mathematics?

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Vector Space of real numbers: Learn

Vector Space of real numbers teaches how lines and angles describe shapes and directions.

Core idea:
- A line can be straight, parallel, perpendicular, horizontal, vertical, or slanting.
- An angle is a turn between two lines.
- Draw and label diagrams carefully before measuring or classifying.

Key words:
- Vector Space of real numbers
- Vector
- Space
- real
- numbers
- method
- model
- unit

Vector Space of real numbers: Worked Example

Worked example for Vector Space of real numbers: A shape has one right angle. How can you show it clearly?

1. Draw the two meeting lines.
2. Mark the square corner to show a right angle.
3. Label it 90 degrees.
4. Name a classroom example, such as a book corner.

Answer: A right angle is a quarter turn, 90 degrees.

Vector Space of real numbers: Vocabulary And Study Skill

Use these words and study moves before you practise Vector Space of real numbers. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:
- Vector Space of real numbers
- Vector
- Space
- real
- numbers
- method
- model
- unit

Study skill:
1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Vector Space of real numbers without a clear example, method, or evidence.

Vector Space of real numbers: Guided Activity

Guided activity for Vector Space of real numbers:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Vector Space of real numbers: Practice And Correction

Practice for Vector Space of real numbers:

Main outcome: use concepts of vectors in 2D to solve related problems such as distance, angles,….

1. Draw one right angle and label it 90 degrees.
2. Name one pair of parallel lines in the classroom.
3. Use a ruler to draw a straight line segment and label its endpoints.

Answer check:
Answer check: the right angle should show a square corner; parallel lines stay the same distance apart.

Correction check:
- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to vector space of real numbers and solve it.

Vector Space of real numbers: Challenge And Remediation

Use this page after practice.

If you need support:
1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Vector Space of real numbers.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:
1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:
- method is clear
- working is shown step by step
- answer is checked in context

Vector Space of real numbers: Outcome Check

Outcome check for Vector Space of real numbers: Use concepts of vectors in 2D to solve related problems such as distance, angles,….

1. Draw and label one right angle.
2. Name one pair of parallel lines.
3. Explain how you know.

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Matrices of and determinants of order: Lesson Opener

Learning focus: Matrices of and determinants of order

Country link: a Rwandan school, market, farming, building, transport, or data problem for Matrices of and determinants of order.

Progression focus: prepare for assessment and real-world use through extended tasks and correction; show methods, units, checks, and a final explanation.

By the end of this topic, you should be able to:
- use matrices and determinants of order 2 to solve other related problems such as organisation of data in a shopping, in Cryptography, in Physics (problems about quantum or circuits), …

Inquiry questions:
- How can matrices of and determinants of order help you solve a real problem in Subsidiary Mathematics?

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Matrices of and determinants of order: Learn

Matrices of and determinants of order teaches how to collect, organize, read, and explain information.

Core idea:
- Data must answer a clear question.
- Tables, tally charts, pictographs, and bar charts make data easier to read.
- A good conclusion says what the data shows.

Key words:
- Matrices of and determinants of order
- Matrices
- determinants
- order
- method
- model
- unit
- answer

Matrices of and determinants of order: Worked Example

Worked example for Matrices of and determinants of order: A class records favourite fruits: mango 8, banana 12, pineapple 6. Which fruit is most popular?

1. Read the table values.
2. Compare 8, 12, and 6.
3. 12 is the greatest number.

Answer: Banana is most popular.

Matrices of and determinants of order: Vocabulary And Study Skill

Use these words and study moves before you practise Matrices of and determinants of order. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:
- Matrices of and determinants of order
- Matrices
- determinants
- order
- method
- model
- unit
- answer

Study skill:
1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Matrices of and determinants of order without a clear example, method, or evidence.

Matrices of and determinants of order: Guided Activity

Guided activity for Matrices of and determinants of order:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Matrices of and determinants of order: Practice And Correction

Practice for Matrices of and determinants of order:

Main outcome: use matrices and determinants of order 2 to solve other related problems such as organisation of data in a shopping, in Cryptography, in Physics (problems about quantum or circuits), ….

1. Ask 10 learners a simple question and make a tally table.
2. Draw a bar chart from the tally table.
3. Write one conclusion from the data.

Answer check:
Answer check: totals in the tally table and bar chart should match.

Correction check:
- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to matrices of and determinants of order and solve it.

Matrices of and determinants of order: Challenge And Remediation

Use this page after practice.

If you need support:
1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Matrices of and determinants of order.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:
1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:
- method is clear
- working is shown step by step
- answer is checked in context

Matrices of and determinants of order: Outcome Check

Outcome check for Matrices of and determinants of order: Use matrices and determinants of order 2 to solve other related problems such as organisation of data in a shopping, in Cryptography, in Physics (problems about quantum or circuits), ….

1. Write one clear example from the topic.
2. Solve it with labelled steps.
3. Check the answer in context.

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Measures of dispersion: Lesson Opener

Learning focus: Measures of dispersion

Country link: a Rwandan school, market, farming, building, transport, or data problem for Measures of dispersion.

Progression focus: prepare for assessment and real-world use through extended tasks and correction; show methods, units, checks, and a final explanation.

By the end of this topic, you should be able to:
- solve and explain problems about measures of dispersion

Inquiry questions:
- How can measures of dispersion help you solve a real problem in Subsidiary Mathematics?

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Measures of dispersion: Learn

Measures of dispersion teaches a mathematical idea by using a clear model, a labelled representation, step-by-step working, and a final check.

Key words:
- Measures of dispersion
- Measures
- dispersion
- method
- model
- unit
- answer
- check

Measures of dispersion: Worked Example

Worked example for Measures of dispersion: Read the task, choose a representation, solve one step at a time, and check the answer in context.

Model:
1. Name what is known.
2. Choose a diagram, table, equation, or model.
3. Complete the calculation or reasoning.
4. Check the result with estimation or inverse reasoning.

Use a Rwanda classroom, market, garden, transport, or home example.

Measures of dispersion: Vocabulary And Study Skill

Use these words and study moves before you practise Measures of dispersion. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:
- Measures of dispersion
- Measures
- dispersion
- method
- model
- unit
- answer
- check

Study skill:
1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Measures of dispersion without a clear example, method, or evidence.

Measures of dispersion: Guided Activity

Guided activity for Measures of dispersion:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Measures of dispersion: Practice And Correction

Practice for Measures of dispersion:

Main outcome: solve and explain problems about measures of dispersion.

1. Solve one measures of dispersion question using labelled steps.
2. Show your method with a drawing, table, number sentence, or diagram.
3. Check your answer and write one sentence explaining it.

Answer check:
Your answer should show a clear model, correct method, correct labels, and a final sentence.

Correction check:
- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to measures of dispersion and solve it.

Measures of dispersion: Challenge And Remediation

Use this page after practice.

If you need support:
1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Measures of dispersion.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:
1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:
- method is clear
- working is shown step by step
- answer is checked in context

Measures of dispersion: Outcome Check

Outcome check for Measures of dispersion: Solve and explain problems about measures of dispersion.

1. Write one clear example from the topic.
2. Solve it with labelled steps.
3. Check the answer in context.

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Elementary probability: Lesson Opener

Learning focus: Elementary probability

Country link: a Rwandan school, market, farming, building, transport, or data problem for Elementary probability.

Progression focus: prepare for assessment and real-world use through extended tasks and correction; show methods, units, checks, and a final explanation.

By the end of this topic, you should be able to:
- use combinations and permutations to determine probabilities of occurrence of an event

Inquiry questions:
- How can elementary probability help you solve a real problem in Subsidiary Mathematics?

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Elementary probability: Learn

Elementary probability teaches how likely an event is.

Core idea:
- Some events are certain, likely, unlikely, or impossible.
- A simple experiment can help you compare chances.
- Record results before making a conclusion.

Key words:
- Elementary probability
- Elementary
- probability
- method
- model
- unit
- answer
- check

Elementary probability: Worked Example

Worked example for Elementary probability: A bag has 4 red counters and 1 blue counter. Which colour is more likely to be picked?

1. Compare the number of counters.
2. Red has 4 chances; blue has 1 chance.
3. More red counters means red is more likely.

Answer: Red is more likely.

Elementary probability: Vocabulary And Study Skill

Use these words and study moves before you practise Elementary probability. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:
- Elementary probability
- Elementary
- probability
- method
- model
- unit
- answer
- check

Study skill:
1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Elementary probability without a clear example, method, or evidence.

Elementary probability: Guided Activity

Guided activity for Elementary probability:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Elementary probability: Practice And Correction

Practice for Elementary probability:

Main outcome: use combinations and permutations to determine probabilities of occurrence of an event.

1. Write one certain event, one likely event, and one impossible event.
2. Toss a coin 10 times and record heads/tails.
3. Use your results to say what happened more often.

Answer check:
Answer check: certain means must happen; impossible means cannot happen.

Correction check:
- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to elementary probability and solve it.

Elementary probability: Challenge And Remediation

Use this page after practice.

If you need support:
1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Elementary probability.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:
1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:
- method is clear
- working is shown step by step
- answer is checked in context

Elementary probability: Outcome Check

Outcome check for Elementary probability: Use combinations and permutations to determine probabilities of occurrence of an event.

1. Write one clear example from the topic.
2. Solve it with labelled steps.
3. Check the answer in context.

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Fundamentals of trigonometry: Review Clinic 1

Use this review clinic to strengthen Subsidiary Mathematics without copying earlier answers.

1. Choose one idea from Fundamentals of trigonometry that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Rwanda.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

SET OF REAL NUMBERS: Review Clinic 2

Use this review clinic to strengthen Subsidiary Mathematics without copying earlier answers.

1. Choose one idea from SET OF REAL NUMBERS that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Rwanda.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

Linear, Quadratic equations and inequalities: Review Clinic 3

Use this review clinic to strengthen Subsidiary Mathematics without copying earlier answers.

1. Choose one idea from Linear, Quadratic equations and inequalities that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Rwanda.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

Polynomial, Rational and Irrational functions: Review Clinic 4

Use this review clinic to strengthen Subsidiary Mathematics without copying earlier answers.

1. Choose one idea from Polynomial, Rational and Irrational functions that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Rwanda.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

Glossary

Use this glossary to revise important words in Subsidiary Mathematics.

- Fundamentals of trigonometry: Explain this term using an example from the book, your school, or your community.
- SET OF REAL NUMBERS: Explain this term using an example from the book, your school, or your community.
- Linear, Quadratic equations and inequalities: Explain this term using an example from the book, your school, or your community.
- Polynomial, Rational and Irrational functions: Explain this term using an example from the book, your school, or your community.
- Limits of polynomial, rational and irrational functions: Explain this term using an example from the book, your school, or your community.
- Differentiation of polynomials, rational and irrational functions and their applications: Explain this term using an example from the book, your school, or your community.
- Vector Space of real numbers: Explain this term using an example from the book, your school, or your community.
- Matrices of and determinants of order: Explain this term using an example from the book, your school, or your community.
- Measures of dispersion: Explain this term using an example from the book, your school, or your community.
- Elementary probability: Explain this term using an example from the book, your school, or your community.

Add five more words from your lessons and write your own meanings.

Answer And Teacher Notes

A strong Subsidiary Mathematics answer shows the model, method, units or labels, final answer, and a check. Use the worked examples and answer checks in each practice page to correct your work. When an answer is wrong, find the first step that changed the meaning and correct that step before trying a new question.

Final Project

Create a final project for Subsidiary Mathematics that shows what you can now do independently.

Your project must include:
1. A clear title.
2. A real-life problem, question, text, performance, investigation, design, or worked task.
3. Evidence from at least three lessons in this book.
4. A drawing, table, map, chart, model, dialogue, paragraph, performance plan, practical product, or worked solution.
5. A correction note showing one improvement after feedback.
6. A short reflection explaining what you learned, what was difficult, and what you will practise next.

Presentation check: keep the work neat, label important parts, use correct vocabulary, and be ready to explain your choices to a classmate or teacher.